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The Neuroscience of Everybody's Favorite Topic

Why do people spend so much time talking about themselves?

By Adrian F. Ward on July 16, 2013

Human beings are social animals. We spend large portions of our waking hours communicating with others, and the possibilities for conversation are seemingly endless—we can make plans and crack jokes; reminisce about the past and dream about the future; share ideas and spread information. This ability to communicate—with almost anyone, about almost anything—has played a central role in our species' ability to not just survive, but flourish.

How do you choose to use this immensely powerful tool—communication? Do your conversations serve as doorways to new ideas and experiences? Do they serve as tools for solving the problems of disease and famine?

Or do you mostly just like to talk about yourself?

If you're like most people, your own thoughts and experiences may be your favorite topic of conversation. On average, people spend 60 percent of conversations talking about themselves—and this figure jumps to 80 percent when communicating via social media platforms such as Twitter or Facebook.

Why, in a world full of ideas to discover, develop, and discuss, do people spend the majority of their time talking about themselves? Recent research suggests a simple explanation: *because it feels good*.

In order to investigate the possibility that self-disclosure is intrinsically rewarding, researchers from the Harvard University Social Cognitive and Affective Neuroscience Lab utilized functional magnetic resonance imaging (fMRI). This research tool highlights relative levels of activity in various neural regions by tracking changes in blood flow; by pairing fMRI output with behavioral data, researchers can gain insight into the relationships between

behavior and neural activity. In this case, they were interested in whether talking about the self would correspond with increased neural activity in areas of the brain associated with motivation and reward.

In an initial fMRI experiment, the researchers asked 195 participants to discuss both their own opinions and personality traits and the opinions and traits of others, then looked for differences in neural activation between self-focused and other-focused answers. Because the same participants discussed the same topics in relation to both themselves and others, researchers were able to use the resulting data to directly compare neural activation during self-disclosure to activation during other-focused communication.

Three neural regions stood out. Unsurprisingly, and in line with previous research, self-disclosure resulted in relatively higher levels of activation in areas of the medial prefrontal cortex (MPFC) generally associated with self-related thought. The two remaining regions identified by this experiment, however, had never before been associated with thinking about the self: the nucleus accumbens (NAcc) and the ventral tegmental area (VTA), both parts of the mesolimbic dopamine system.

These newly implicated areas of the brain are generally associated with reward, and have been linked to the pleasurable feelings and motivational states associated with stimuli such as sex, cocaine, and good food. Activation of this system when discussing the self suggests that self-disclosure, like other more traditionally recognized stimuli, may be inherently pleasurable—and that people may be motivated to talk about themselves more than other topics (no matter how interesting or important these non-self topics may be).

This experiment left at least one question unanswered, however. Although participants were revealing information about themselves, it was unclear whether or not anyone was paying attention; they were essentially talking without knowing who (if anyone) was on the other end of the line. Thus, the reward- and motivation-related neural responses ostensibly produced by self-disclosure *could* be produced by the act of disclosure—of revealing information about the self to someone else—but they could also be a result of focusing on the self more generally—whether or not anyone was listening.

In order to distinguish between these two possibilities, the researchers conducted a follow-up experiment. In this experiment, participants were asked to bring a friend or relative of their choosing to the lab with them; these companions were asked to wait in an adjoining room while participants answered questions in a fMRI machine. As in the first study, participants responded to questions about either their own opinions and attitudes or the opinions and attitudes of someone else; unlike in the first study, these participants were explicitly told whether their responses would be “shared” or “private”; shared responses were relayed in real time to each participant’s companion and private responses were never seen by anyone, including the researchers.

In this study, answering questions about the self always resulted in greater activation of neural regions associated with motivation and reward (i.e., NAcc, VTA) than answering questions about others, and answering questions publicly always resulted in greater activation of these areas than answering questions privately. Importantly, these effects were additive; both talking about the self and talking to someone else were associated with reward, and doing both produced greater activation in reward-related neural regions than doing either separately.

These results suggest that self-disclosure—revealing personal information to others—produces the highest level of activation in neural regions associated with motivation and reward, but that introspection—thinking or talking about the self, in the absence of an audience—also produces a noticeable surge of neural activity in these regions. Talking about the self is intrinsically rewarding, even if no one is listening.

Talking about the self is not at odds with the adaptive functions of communication. Disclosing private information to others can increase interpersonal liking and aid in the formation of new social bonds—outcomes that influence everything from physical survival to subjective happiness. Talking about one's own thoughts and self-perceptions can lead to personal growth via external feedback. And sharing information gained through personal experiences can lead to performance advantages by enabling teamwork and shared responsibility for memory. Self-disclosure can have positive effects on everything from the most basic of needs—physical survival—to personal growth through enhanced self-knowledge; self-disclosure, like other forms of communication, seems to be adaptive.

You may like to talk about yourself simply because it feels good—because self-disclosure produces a burst of activity in neural regions associated with pleasure, motivation, and reward. But, in this case, feeling good may be no more than a means to an end—it may be the immediate reward that jump-starts a cycle of self-sharing, ultimately leading to wide varieties of long-term benefits.

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ABOUT THE AUTHOR(S)

Adrian F. Ward is postdoctoral research associate in the Leeds School of Business at the University of Colorado, Boulder. He received his doctorate from the Department of Psychology at Harvard University. His research is focused on technology, cognition, social relationships, and decision-making. Follow him @adrianfward

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